

Build advanced and highly reliable industrial Ethernet communication system



ISM7312D-P Managed Industrial Ethernet Switch

· PoE 802.3af/at

· Layer 2+ Managed

· STP/RSTP/MSTP

· IP30 Fanless

1. Product Overview

The ISM series of managed industrial Ethernet switches is specifically designed to meet the demands of harsh industrial environments such as power, transportation, and industrial control applications. The rugged casing design and protected circuits among other industrial-grade features perform its remarkable adaption to rough environments, reliably operating in extreme temperature conditions.

Product Features:

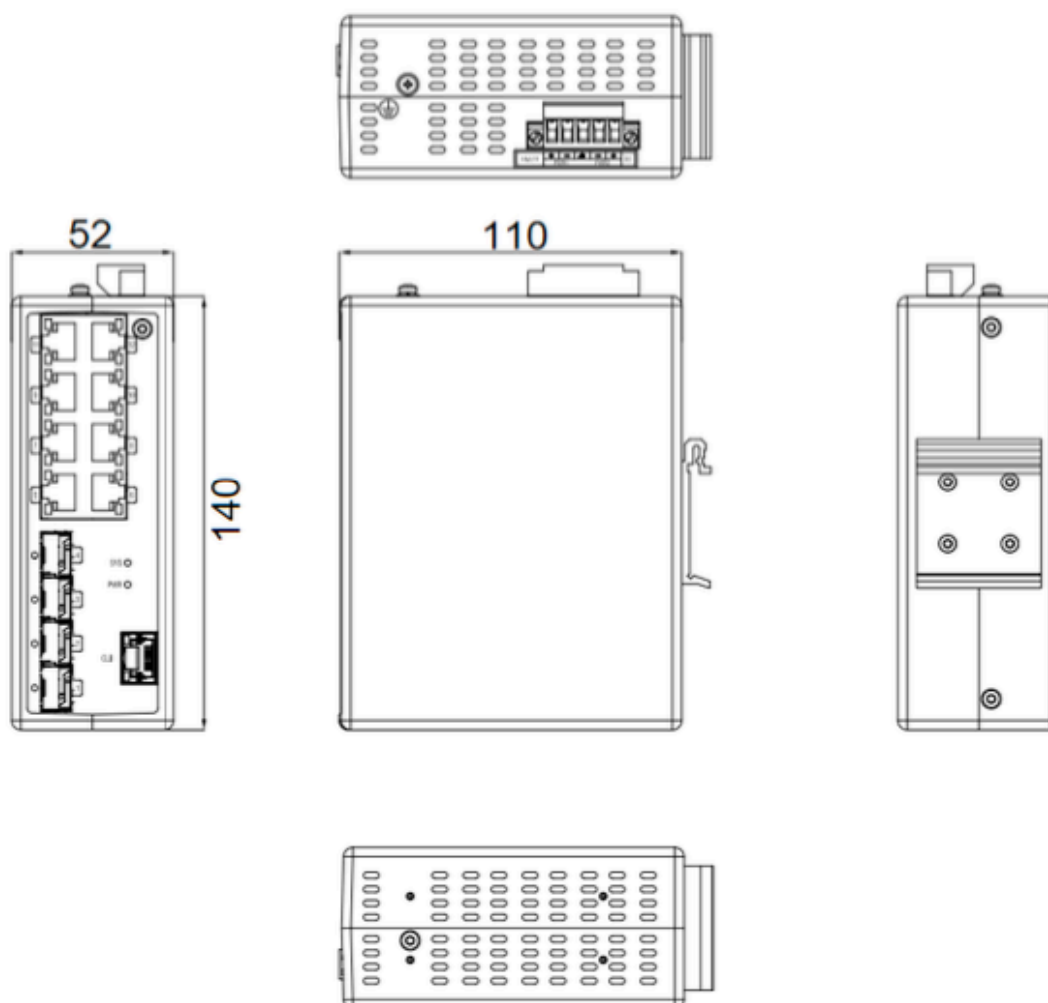
- **Reliability & Resilience:** Fanless design, IP30 protection rating, robust metal casing with protective coating, dust and dirt resistance, and wide temperature operation (-40 °C ~ +75 °C (-40 °F ~ +167 °F)). Industrial-grade redundant dual input power supply design. MTBF over 35 years.
- **PoE Power Supply:** Supports PoE (Power over Ethernet), compatible with IEEE 802.3af/at standards. Each PoE port provides up to 30 W output power across 8 Gigabit ports.
- **Redundancy Protocols:** Supports STP/RSTP/MSTP ring redundancy protocols, offering flexible choices for constructing intricate industrial Ethernet communication systems.
- **Network Management:** Supports SNMPv1/v2c/v3 for integrated network management and RMON for effective monitoring and fault prediction. DIN-rail installation, small size, plug-and-play for quick deployment.
- **Enhanced Security & Control:** Robust network security with IGMP Snooping, VLAN network segmentation (port-based, IEEE 802.1Q VLAN, GVRP), and QoS (IEEE 802.1P/1Q and ToS/DiffServ) to increase determinism.

Key Technical Specifications

Parameter	Specification
Type	Layer 2+ Managed Industrial Ethernet Switch
Ports	8 × 10/100/1000BaseT PoE + 4 × 1000BaseX SFP + 1 × Management Serial CLI

Parameter	Specification
PoE	IEEE 802.3af/at, up to 30 W per port (8 ports)
Switching Performance	68 Gbps backplane bandwidth; 16K MAC table; 4 Mbit buffer
Dimensions / Weight	52 × 140 × 110 mm (2.05 × 5.51 × 4.33 in) / 0.7 kg (1.54 lb)
Power	48–54V DC redundant dual inputs
Environment	-40 °C ~ +75 °C (-40 °F ~ +167 °F) operating; IP30; fanless
Management	Web, CLI, SNMPv1/v2c/v3, RMON
Protocols	STP/RSTP/MSTP, IGMP Snooping, VLAN, GVRP, QoS
EMC / Certifications	FCC Part 15 Class A; EN55022 / EN55035 Class A; IEC 61000-4 series

2. Product Dimensions



ISM7312D (52 × 140 × 110 mm (2.05 × 5.51 × 4.33 in))

Note:

1. All dimensions are in millimeters (mm), with inches (in) in parentheses.
2. All dimensions are approximate and **for reference only**.
3. The dimensions shown in the figure **shall not be used for production or processing**.
4. Dimensions must comply with part and manufacturing tolerance requirements.
5. Dimensions are subject to change without notice.

3. Technical Specifications

3.1 Protocol Compliance List

Category/Parameter	Specification
IEEE Standards	
IEEE 802.3	CSMA/CD method and physical Layer specifications
IEEE 802.1p	Priority Queuing
IEEE 802.1q	VLAN tagging
IEEE 802.1d	Spanning Tree Algorithm
IEEE 802.1w	Rapid Spanning Tree
IEEE 802.1s	Multiple Spanning Tree
IEEE 802.3ac	VLAN Tagging
IEEE 802.1x	Authentication
IEEE 802.3ad	Link Aggregation
IEEE 802.3x	Flow Control
IEEE 802.3	Ethernet
IEEE 802.3u	Fast Ethernet
IEEE 802.3z	Gigabit Ethernet
IEEE 802	Networks
RFC Standards	
RFC 213	DHCP Server

Category/Parameter	Specification
RFC 768	UDP
RFC 791	IP
RFC 792	ICMP
RFC 793	TCP
RFC 826	ARP
RFC 854	Telnet Client & Server
RFC 862	Echo Protocol
RFC 863	Discard Protocol
RFC 904	Exterior Gateway Protocol Formal Specification
RFC 1027	Using ARP to Implement Transparent Subnet Gateways
RFC 1058	RIP
RFC 1059, 1119	NTPv1/2
RFC 1112	IGMP
RFC 1191	Path MTU Discovery
RFC 1256	ICMP Router Discovery Protocol
RFC 1267	A Border Gateway Protocol 3 (BGP-3)
RFC 1388	RIP Version 2 Carrying Additional Information
RFC 1403	BGP OSPF Interaction
RFC 1519	CIDR (Classless Inter-domain Routing)
RFC 1587	OSPF NSSA
RFC 1812	Requirements for IP Version 4 Routers

Category/Parameter	Specification
RFC 1994	PPP Challenge Handshake Authentication Protocol (CHAP)
RFC 2068	HTTP
RFC 2138	RADIUS
RFC 2139	RADIUS Accounting
RFC 2236	IGMPv2
RFC 2328	OSPFv2
RFC 2338	VRRP
RFC 2370	The OSPF Opaque LSA Option
RFC 2474	DiffServ Precedence
RFC 2475	DiffServ Core and Edge Router Functions
RFC 2597	DiffServ Assured Forwarding
RFC 2598	DiffServ Expedited Forwarding
RFC 2644	Directed Broadcasts
RFC 2865	Remote Authentication Dial In User Service (RADIUS)
RFC 3046	DHCP Relay Agent Information Option
RFC 3222	Forwarding Information Base (FIB)
Other Protocols	
GMRP	GARP
GVRP	GARP
SSH2	Secure Shell 2
IGMP	Snooping

Category/Parameter	Specification
SNMPv3	Supported

3.2 Hardware Specifications

Category/Parameter	Specification
Physical Performance	
Enclosure	Fully enclosed seamless metal enclosure
Dimensions (W × D × H)	52 mm × 140 mm × 110 mm
Weight	0.7 kg (1.54 lb)
Mounting Method	DIN-rail mounting
Cooling Method	Fanless cooling
Ingress Protection	IP30
Storage Temperature	-40 °C ~ +85 °C (-40 °F ~ +185 °F)
Operating Temperature	-40 °C ~ +75 °C (-40 °F ~ +167 °F)
Humidity	5 ~ 95% (non-condensing)
Hardware Performance	
Backplane Bandwidth	68 Gbps
Transmission Mode	Store-and-Forward
MAC Table Size	16K

Category/Parameter	Specification
Packet Buffer Size	4 Mbits
Exchange Rate	148,800 pps/100M ports; 1,488,000 pps/1000M ports
Power Parameters	
Input Voltage	48~54V DC, redundant dual input
Overload Current Protection	Supported
Reverse Polarity Protection	Supported
PoE	
PoE 10/100/1000BaseT Ports	8 ports
Maximum Power	15.4 W (IEEE 802.3af); 30 W (IEEE 802.3at)
Electromagnetic Characteristics	
EMI	FCC 47 CFR Part 15 Class A; EN55022 Class A; EN55035 Class A
EMS	IEC(EN)61000-4-2, Class 4 IEC(EN)61000-4-3, Class 3 IEC(EN)61000-4-4, Class 4 IEC(EN)61000-4-5, Class 4 IEC(EN)61000-4-6, Class 3 IEC(EN)61000-4-8, Class 5
Mechanical Characteristics	
Shock	IEC60068-2-27

Category/Parameter	Specification
Freefall	IEC60068-2-31
Vibration	IEC60068-2-6
Quality Assurance	
Warranty Period	5 years
MTBF	35 years

4. Software Specifications

Software Functions	Specification
Redundancy	STP, MSTP, RSTP, Port Trunking
Management Mode	Browser, Serial Port, STD-17 MIB-II, STD-58 SMIV2, STD-59 RMON, STD-62 SNMPv3, SNMPv2c, SNMPv1, RFC2925 Ping MIB
Time Synchronization	SNTP
Diagnostic Mode	Indicator light, Journal File, RMON, Port Mirroring, TRAP
Others	IGMP Snooping, VLAN segmentation, QoS (IEEE 802.1P/1Q, ToS/DiffServ), GVRP, multicast filtering

5. Ordering Guide

Model	Description
ISM7312D-P-4GSFP-8GT-48	12-Port Layer 2+ Managed Industrial Switch. 4 × 1000BaseX SFP Ports (SFP module not included), 8 × 802.3af/at PoE 10/100/1000BaseT Ports. 1 Management Serial CLI Port, IP30 Protection Class, Operating Temperature from -40 °C to +75 °C (-40 °F to +167 °F). Dual 48-54VDC Power Inputs. CLI and power cables are ordered separately.

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